LEAH NEWMAN

BIOMEDICAL ENGINEERING STUDENT AT TRINITY COLLEGE DUBLIN | FOUNDATION SCHOLAR | NAUGHTON SCHOLAR

Third-year Biomedical Engineering student at Trinity College Dublin, Foundation Scholar and Naughton Scholar, with interests in robotics, electronics, medical devices and computational engineering. Experience spans embedded systems, hardware prototyping and neurological disability research, including wireless debugging tools and bespoke 3D-printed assistive devices.

EXPERIENCE

PATCH ASSOCIATE AND GRANTS LEAD

May 2026 – Present, Dogpatch Labs

- Supported the 2026 Patch accelerator through participant mentoring, workshops, hackathons and Demo Day.
- Lead technical project reviews for Patch Grants, assessing feasibility, budgets and development plans alongside the grants team.
- Interview applicants, provide technical feedback and connect successful grantees with relevant mentors.

STUDENT RESEARCHER – NEUROLOGICAL DISABILITY AND ASSISTIVE TECHNOLOGY

May 2025 – Present, Beaumont Hospital, TCD, DCU

- Conducted research across two summers into bespoke assistive devices and digital technologies for neurological disability care.
- Designed and prototyped 3D-printed assistive devices, collaborating with clinicians and engineers to address individual functional needs.
- Developed a national clinician survey investigating the use of 3D printing in neurological rehabilitation and barriers to adoption.
- Built a scheduling system for multidisciplinary clinics to coordinate appointments and clinician availability, aiming to improve patient flow and reduce waiting times.
- Presented assistive-device research at MND Global Awareness Day 2025.

CONTACT

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Drogheda / Dublin

RESULTS

Engineering TCD Year 1 - 86%
Engineering TCD Year 2 – 83%

Leaving Certificate 2024:
7 H1s, with 100% in Physics, Technology (Top Nationally), Mathematics and Applied Mathematics

EDUCATION

TRINITY COLLEGE DUBLIN

Integrated Engineering programme, Biomedical Engineering stream – Year 3

- Foundation Scholar.
- Naughton Foundation Scholar, 2024.
- Relevant study: programming, electronics, signals and systems, mechanics and materials.

ST OLIVER'S COMMUNITY COLLEGE

Competing at the National level in Mathematics, Science and Technology

625 Points

PATCH SUMMER ACCELERATOR

June 2025 - Aug 2025

- Selected as one of 35 participants nationally for the Patch summer accelerator.
- Co-developed ScopeLab, a prototype wireless debugging tool for robotics systems, within a four-person team.

PROJECTS

WI-FI CONTROLLED ROBOTIC BUGGY

January - April 2026

- Developed Arduino firmware and a Processing interface for wireless control and live monitoring of a robotic buggy.
- Integrated motor control, encoder feedback, ultrasonic obstacle sensing and infrared line sensors.
- Implemented and evaluated PID speed-profile tracking using recorded speed data and error measurements.

SCOPELAB

July 2025 – Aug 2025

- Co-founded ScopeLab, a wireless debugging tool for real-time monitoring of robotics data and system health.
- Led numerical, computational and mechanical development, including signal decoding, ASCII conversion and data displaying.
- Created CAD models for the device enclosure.

OUT OF THIS WORLD ROLLERCOASTERS

Jan 2024 - May 2024

- Modelled rollercoaster motion under different planetary gravitational conditions, accounting for friction and air resistance.
- Received the SciFest Maths in Science Award.

THE MULTITASKER – MODULAR ASSISTIVE DEVICE

May 2021 – July 2024

- A 3D printed tool for people with limited hand mobility, customizable to the users' unique set of needs and struggles.
- Triple-award winning project at regional and national levels.

MAIN AWARDS

- Trinity Foundation Scholar
- Naughton Scholar
- ETTA top LC Technology result and national finalist project
- AMTCE 3D Printing and Engineering Competition:1st place,
- Scifest Boston Scientific Award, Maths in science award, Video outreach award and national finalist.
  Irish mathematics Olympiad Finalist, 7th in Maynooth,
  Finalist in IMTA Junior Mathematics Competition

STEM OUTREACH

SCIFEST JUDGE

Judge at my secondary school's inaugural (and now annual) Scifest at schools' competition, where I had the opportunity to give feedback to students and share my love for STEM and research.

AMTCE JUDGE

Judge at the 3D printing and engineering regional competition that we won in Transition Year

SCIFEST MENTOR

Provide guidance to students in the early stage of their projects for the regional and national rounds.

TECHNICAL SKILLS

Programming: C++. Arduino, Processing

Embedded Sysytems: Arduino, Sensor Integration

CAD and Fabrication: Machining, Tools, Solidworks, OnShape, 3D Printing and assembly

Electronics: Ltspice, oscilloscope

Control: PID control, mathematical modelling

REFEREES

DR DEIRDRE MURRAY

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Research Supervisor in Beaumont Hospital and TCD
Clinical Specialist Physiotherapist, Researcher and Assoc. Prof. DCU

PROF ORLA HARDIMAN

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Consultant Neurologist and head of the School of Neurology, Trinity College Dublin

DR SANDRA COLLEARY

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Mathematics Teacher from 2019 to 2024 and tutor